

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2023/2024

OCTOBER EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

MONDAY, 23 OCTOBER 2023

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) Artificial intelligence (A.I.) is a branch of computer science that focuses on building intelligent machines that function and respond just like people. There are four approaches to A.I. where they vary along two dimensions that are concerned with 1) thought vs. behaviour, and 2) human performance vs. rationality. For instance, one of them is the Turing test approach that intends to develop A.I. systems that are capable of acting like a human.
- (i) Name the other **THREE (3)** approaches and briefly describe each of them. (3 + 3 marks)
 - (ii) Draw a table with the use of labels to illustrate how these **FOUR (4)** approaches differ from each other based on the two dimensions. (6 marks)
- b) *ChatGPT* is a revolutionary new A.I. chat-bot that creates human-like conversational dialogue. Its achievement has surpassed many users' expectations and it already has more than 100 million active users two months after its first launch at the end of 2022. The *Turing test* is a test of a machine's (i.e., especially chat-bots) ability to exhibit intelligent behaviour equivalent to, or indistinguishable from, that of a human. There are many variants of this test and the *total Turing test* is one of them.
- (i) Identify how the *total Turing test* is different from the *Turing test*. (2 marks)
 - (ii) Identify whether *ChatGPT* can pass the *Turing test*. Justify your answer by focusing on the capabilities of *ChatGPT*. (1 + 4 marks)
 - (iii) Identify whether *ChatGPT* can pass the *total Turing test*. Justify your answer by focusing on the capabilities of *ChatGPT*. (1 + 2 marks)
 - (iv) Which one is more relevant to the definition of A.I. provided in **Question 1 a)**, the *Turing test* or the *total Turing test*? Justify your answer. (1 + 2 marks)

[Total: 25 marks]

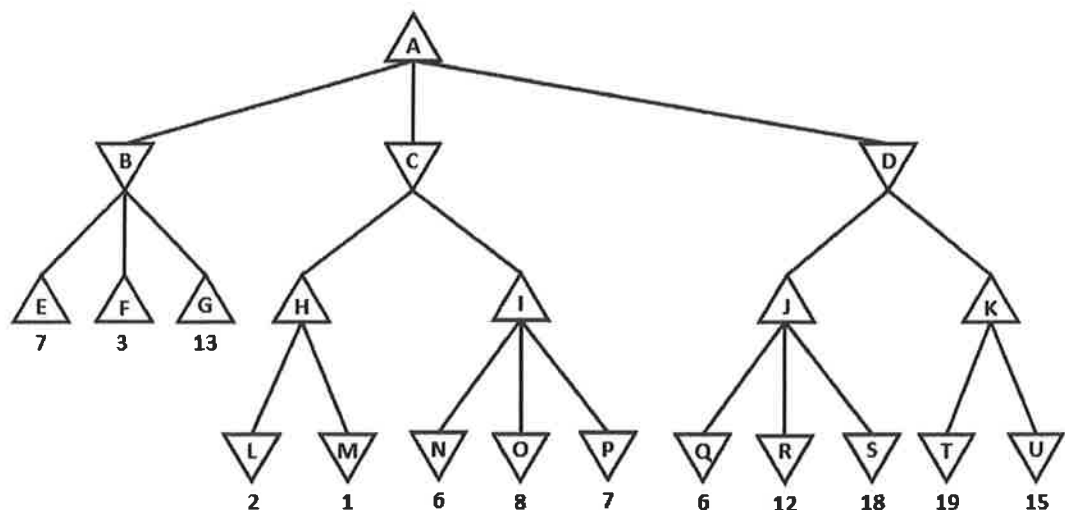
Question 2

- a) The world's first self-driving taxi service has been launched in Singapore in 2016. The *PEAS* description can be used to describe the problems to which the self-driving taxi solves.
- (i) Identify what the acronym *PEAS* stands for. (4 marks)
 - (ii) Construct a *PEAS* description for this problem by providing **ONE (1)** example for each of the four elements. (4 marks)
 - (iii) Identify **SEVEN (7)** properties of the environment. Justify your answer. (7 + 7 marks)
 - (iv) Identify whether the self-driving taxi is a rational agent or an omniscient agent. Justify your answer. (1 + 2 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

- a) In A.I., search techniques are universal problem-solving methods. Rational agents or problem-solving agents in A.I. mostly used these search strategies or algorithms to solve a specific problem and provide the best result. *Greedy-Best First (GBF)* and *Steepest-Ascent Hill Climbing (SAHC)* are two examples of popular search techniques.
- Briefly describe *GBF* and *SAHC*. (2 marks)
 - Identify **TWO (2)** commonalities between *GBF* and *SAHC*. (2 marks)
 - Discuss **TWO (2)** differences between *GBF* and *SAHC*. (4 marks)
 - Suppose you are driving in a city. Identify whether *GBF* or *SAHC* is more appropriate to be deployed in solving the problem of searching for a route from one location to another. Justify your answer. (1 + 2 marks)
- b) Search techniques are also used to solve games such as Tic-Tac-Toe and chess. For instance, the minimax algorithm is one popular technique used to explore a game tree (which is created by each move made by two players, MAX and MIN, alternatively within a game) in search of the optimal move for a player.

**Figure 1. Game tree**

Answer the following three questions based on the game tree provided in **Figure 1**.

- Compute the minimax value for nodes A, B, C, D, H, I, J, and K. (8 marks)
- Identify the best move for the MAX player at the root node. (1 mark)
- Apply minimax search to the game tree with alpha-beta pruning in left-to-right order to deduce nodes that would be pruned. (5 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

- a) Knowledge-based agents are a type of agents that is capable of maintaining an internal state of knowledge, reasoning over that knowledge and taking actions. Knowledge representation can be done with the use of propositional logic. For instance, atomic sentences and complex sentences are two types of propositions that can be represented through propositional logic.

Table 1: Propositions and their symbols

Symbol	Proposition
D	The sky is dark.
C	The sky is cloudy.
R	It is going to rain.
M	Mark is happy.
T	The trip is postponed.
S	Sandy is sleeping.

Referring to the information given in **Table 1**, translate the following English sentences into propositions with the use of symbols and logical connectives (if applicable).

- (i) It is going to rain. (1 mark)
 - (ii) Sandy is not sleeping. (1 mark)
 - (iii) The sky is dark and cloudy. (1 mark)
 - (iv) The sky is dark but it is not going to rain. (1 mark)
 - (v) The trip is postponed or not postponed. (1 mark)
 - (vi) Sandy is sleeping and Mark is not happy. (1 mark)
 - (vii) If the sky is dark or cloudy, then it is going to rain. (1 mark)
 - (viii) If the sky is neither dark nor cloudy, then the trip is not postponed. (1 mark)
 - (ix) It is not the case that if Sandy is sleeping, then Mark is not happy. (1 mark)
 - (x) The trip is postponed if and only if Mark is not happy and the sky is dark. (1 mark)
- b) A proposition is a declarative sentence that is either true or false, but not both. A truth table is a complete list of possible truth values of a given proposition. Construct a truth table for each of the following propositions and identify if it is a tautology.
- (i) $[(P \Rightarrow Q) \wedge P] \Rightarrow P$ (4 marks)
 - (ii) $[\neg P \wedge (P \Rightarrow Q)] \Rightarrow \neg P$ (6 marks)
 - (iii) $(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow P)$ (5 marks)

[Total: 25 marks]